

OPERATING SYSTEMS PROJECT

CPU Scheduling Simulation System

A practical simulation of CPU scheduling algorithms



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Why CPU Scheduling?

The Basic Idea

The CPU can execute only one process at a time. When several processes are ready, the operating system must decide which process gets the CPU next.

01 Multiple Processes



02 Scheduling Decision



03 CPU Execution

What Our System Does

- ▶ Accepts process details from the user.
- ▶ Applies a selected scheduling algorithm.
- ▶ Calculates important scheduling times.
- ▶ Displays execution order using a Gantt Chart.
- ▶ Helps students understand scheduling practically.

Main Inputs

Process ID
Burst Time

Arrival Time
Priority

Project Objectives

01 Simulate Scheduling

Implement major CPU scheduling

02 Calculate Metrics

Find completion, turnaround and waiting times.

03 Visualize Execution

Show process execution using a

04 Support Learning

Make scheduling concepts easier to understand.

Goal:

Turn theoretical CPU scheduling calculations into an interactive practical simulation.

Scheduling Algorithms

FCFS

First Come First Serve

SJF

Shortest Job First

The process with the shortest burst time is selected first.

Priority

Priority Scheduling

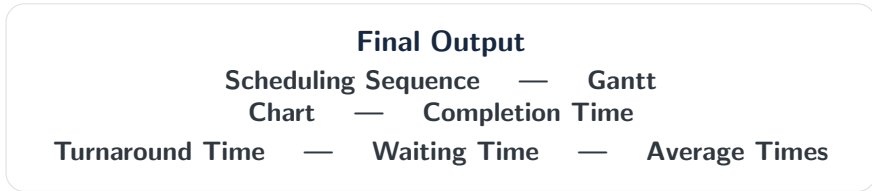
RR

Round Robin

Each process receives CPU time according to the selected time quantum.

The system supports preemptive and non-preemptive scheduling modes where applicable.

System Workflow



Input → Scheduling Logic → Calculations → Visualization

Technology Stack

C Programming

- ▶ Scheduling logic
- ▶ Algorithm implementation
- ▶ Time calculations

HTML CSS

- ▶ Web page structure
- ▶ User interface
- ▶ Visual presentation

JavaScript

- ▶ User interaction
- ▶ Input handling
- ▶ Gantt Chart display

System Architecture

User Interface → JavaScript Interaction →
C Scheduling Logic → Calculated Results

Performance Metrics Gantt Chart

Calculated Metrics

Completion Time Time when

Average Times Average wait-
ing and turnaround time.

Example Gantt Chart



The Gantt Chart provides a visual representation of the order and duration of process execution.

Makes algorithm behaviour easier to compare and understand.

Conclusion Future Scope

Conclusion

The CPU Scheduling Simulation System provides an interactive way to understand how an operating system selects and executes processes.

- ▶ Implements major scheduling algorithms.
- ▶ Calculates important scheduling metrics.
- ▶ Displays execution using a Gantt Chart.
- ▶ Connects theory with practical simulation.

Future Scope

- Add more scheduling algo-
- Support larger process datasets